



FOR OFFICIAL USE

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National  
Qualifications  
2016

Mark

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**X744/75/01****Lifeskills Mathematics  
Paper 1 (Non-Calculator)**

WEDNESDAY, 4 MAY

9:00 AM – 9:50 AM



\* X 7 4 4 7 5 0 1 \*

Fill in these boxes and read what is printed below.

Full name of centre

Town

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Forename(s)

Surname

Number of seat

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Date of birth

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Month

Year

Scottish candidate number

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**Total marks — 35**

Attempt ALL questions.

**You may NOT use a calculator.**

Full credit will be given only to solutions which contain appropriate working.

State the units for your answer where appropriate.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting.

Use **blue** or **black** ink.

Before leaving the examination room you must give this booklet to the Invigilator; if you do not, you may lose all the marks for this paper.



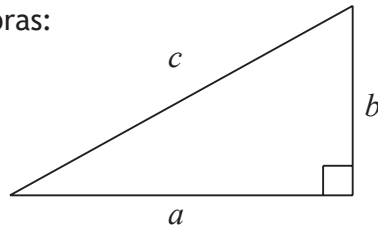
\* X 7 4 4 7 5 0 1 0 1 \*

## FORMULAE LIST

Circumference of a circle:  $C = \pi d$

Area of a circle:  $A = \pi r^2$

Theorem of Pythagoras:



$$a^2 + b^2 = c^2$$

Volume of a cylinder:  $V = \pi r^2 h$

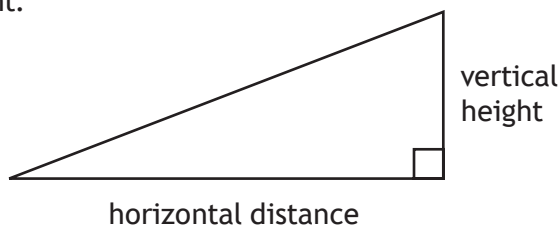
Volume of a prism:  $V = Ah$

Volume of a cone:  $V = \frac{1}{3} \pi r^2 h$

Volume of a sphere:  $V = \frac{4}{3} \pi r^3$

Standard deviation:  $s = \sqrt{\frac{\Sigma(x - \bar{x})^2}{n-1}} = \sqrt{\frac{\Sigma x^2 - (\Sigma x)^2/n}{n-1}}$ , where  $n$  is the sample size.

Gradient:



$$\text{gradient} = \frac{\text{vertical height}}{\text{horizontal distance}}$$



\* X 7 4 4 7 5 0 1 0 2 \*

Total marks — 35  
Attempt ALL questions

1. A restaurant can buy long grain rice in two sizes of bags.

- A 9 kg bag costs £25.65
- A 20 kg bag costs £57.20

Which size of bag is better value for the restaurant?

Use your working to justify your answer.

3

2. Aneesha and Brian are playing a board game. Each move is determined by rolling two dice.



Aneesha requires a total of **10 or more** on her next roll to win the game.

What is the probability of Aneesha winning the game on the next roll?

Give your answer as a fraction.

3

[Turn over



3. Gary lives in Biggar and has to go to a meeting in Edinburgh.  
He plans to travel to his meeting by bus.  
He uses this bus timetable to plan his journey.

| Dumfries ● Biggar ● Edinburgh  |      |      |      |      |      |      |      |      |      |      |      |      |
|--------------------------------|------|------|------|------|------|------|------|------|------|------|------|------|
| Monday to Friday               |      |      |      |      |      |      |      |      |      |      |      |      |
| Route Number                   | 101  | 101  | 101  | 101  | 101  | 102  | 101  | 101  | 101  | 101  | 101  | 102  |
| Dumfries Whitesands Stance 4   | 0535 | 0710 |      |      | 0910 | 1025 |      |      | 1315 |      |      | 1815 |
| Heathhall                      | 0543 | 0720 |      |      | 0920 |      |      |      | 1325 |      |      |      |
| Amisfield Main Rd              | 0547 | 0725 |      |      | 0925 |      |      |      | 1330 |      |      |      |
| Parkgate                       | 0552 | 0730 |      |      | 0930 |      |      |      | 1335 |      |      |      |
| St Ann's                       | 0557 | 0736 |      |      | 0936 |      |      |      | 1341 |      |      |      |
| Beattock Primary School        | 0606 | 0745 |      |      | 0945 |      |      |      | 1350 |      |      |      |
| Moffat High St Stance 2        | 0612 | 0752 |      |      | 0952 |      |      |      | 1357 |      |      |      |
| Holywood                       |      |      |      |      |      | 1031 |      |      |      |      |      | 1821 |
| Auldgirth                      |      |      |      |      |      | 1039 |      |      |      |      |      | 1829 |
| Closeburn                      |      |      |      |      |      | 1046 |      |      |      |      |      | 1836 |
| Thornhill Cross                |      |      |      |      |      | 1050 |      |      |      |      |      | 1840 |
| Durisdiermill                  |      |      |      |      |      | 1100 |      |      |      |      |      | 1850 |
| Troloss                        |      |      |      |      |      | 1105 |      |      |      |      |      | 1855 |
| Elvanfoot                      |      |      |      |      |      | 1117 |      |      |      |      |      | 1907 |
| Crawford                       | 0633 | 0813 |      |      | 1013 | 1128 |      |      | 1418 |      |      | 1913 |
| Abington Village               | 0640 | 0820 |      |      | 1020 | 1135 |      |      | 1425 |      |      | 1920 |
| Abington Service Area          | 0646 | 0827 |      |      | 1027 | 1142 |      |      | 1432 |      |      | 1927 |
| Roberton                       | 0651 | 0832 |      |      | 1032 | 1147 |      |      | 1437 |      |      | 1932 |
| Lamington                      | 0657 | 0838 |      |      | 1038 | 1153 |      |      | 1443 |      |      | 1938 |
| Coulter                        | 0702 | 0843 |      |      | 1043 | 1158 |      |      | 1448 |      |      | 1943 |
| Biggar                         | 0633 | 0709 | 0853 | 0953 | 1053 | 1208 | 1253 | 1353 | 1458 | 1623 | 1803 | 1953 |
| Dolphinton                     | 0644 | 0721 | 0905 | 1004 | 1104 | 1219 | 1304 | 1404 | 1509 | 1634 | 1814 | 2004 |
| West Linton                    | 0651 | 0731 | 0915 | 1011 | 1111 | 1226 | 1311 | 1411 | 1516 | 1641 | 1821 | 2011 |
| Carlops                        | 0655 | 0735 | 0920 | 1015 | 1115 | 1230 | 1315 | 1415 | 1520 | 1645 | 1825 | 2015 |
| Silverburn                     | 0702 | 0741 | 0927 | 1021 | 1121 | 1236 | 1321 | 1421 | 1526 | 1651 | 1831 | 2021 |
| Penicuik Town Centre Stop C    | 0707 |      | 0932 | 1026 |      | 1241 | 1326 | 1426 |      | 1656 | 1836 | 2026 |
| Flotterstone                   | 0717 | 0746 | 0942 | 1036 | 1126 | 1251 | 1336 | 1436 | 1531 | 1706 | 1846 | 2034 |
| Fairmilehead, Swanston Drive   | 0724 | 0753 | 0948 | 1042 | 1132 | 1257 | 1342 | 1442 | 1537 | 1712 | 1852 | 2039 |
| Morningside Station            | 0732 | 0801 | 0956 | 1050 | 1140 | 1305 | 1350 | 1450 | 1545 | 1720 | 1900 | 2045 |
| Tollcross                      | 0740 | 0809 | 1004 | 1058 | 1148 | 1313 | 1358 | 1458 | 1553 | 1728 | 1908 | 2050 |
| Lothian Road, Caledonian Hotel | 0749 | 0818 | 1011 | 1104 | 1154 | 1319 | 1404 | 1504 | 1600 | 1735 | 1914 | 2055 |
| Edinburgh Bus Stance E         | 0801 | 0830 | 1021 | 1114 | 1204 | 1329 | 1414 | 1514 | 1611 | 1745 | 1924 | 2102 |

His meeting in Edinburgh starts at 11:30 am.

It will take him 25 minutes to walk from the Edinburgh bus stance to his meeting.

What is the latest bus he can catch in Biggar to be at his meeting on time?

2



\* X 7 4 4 7 5 0 1 0 4 \*

4. Seonaid is saving up to buy a tablet computer costing £388.  
She earns £7·30 per hour and works for 30 hours each week.  
Seonaid is paid at the end of each week.  
She pays £5·32 in Income Tax and £7·68 in National Insurance each week.  
Her living expenses are £86 per week.  
Seonaid saves **half** of the money that she has left each week towards the tablet computer.  
How many weeks will it take her to save up enough money to buy the computer?

3

[Turn over



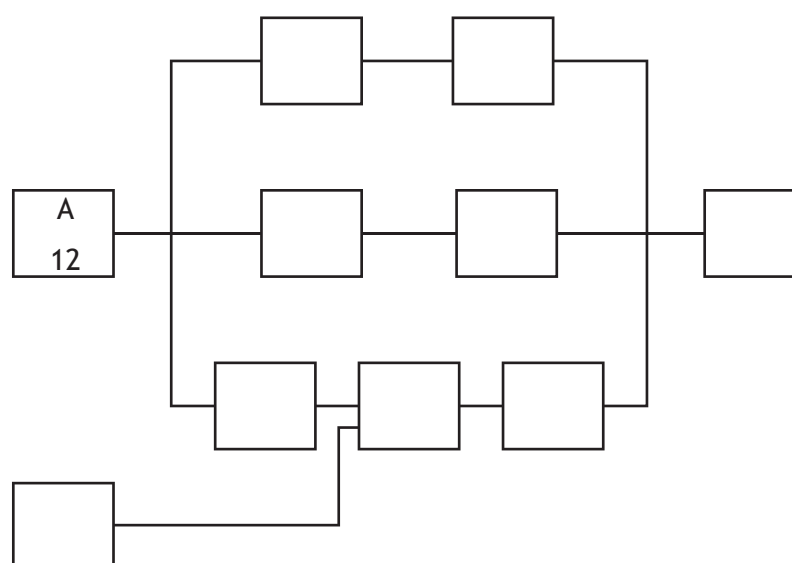
5. A computer company is researching how long it would take to develop a new games console and bring it to market.

The following table of necessary tasks was produced.

| Activity | Description         | Preceding Task | Time (months) |
|----------|---------------------|----------------|---------------|
| A        | Product design      | None           | 12            |
| B        | Market research     | None           | 2             |
| C        | Production analysis | A              | 3             |
| D        | Product model       | A              | 4             |
| E        | Sales brochure      | A              | 1             |
| F        | Product testing     | D              | 5             |
| G        | Cost analysis       | C              | 3             |
| H        | Sales training      | B,E            | 2             |
| I        | Pricing             | H              | 1             |
| J        | Project report      | F,G,I          | 1             |

- (a) Complete the diagram below to show the tasks and times in the boxes.  
(An additional diagram, if required, can be found on *Page 12*).

2

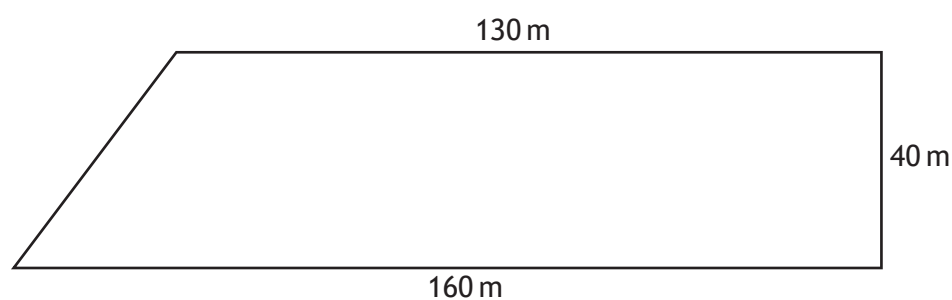


- (b) The company want this entire process to be completed in 2 years.  
Based on the times given, is this possible?  
Show working to justify your answer.

2



6. A farmer needs to **completely enclose** this field with a new fence.



The fence is only sold in 80 metre rolls.

Each roll costs £73.99.

Calculate the cost of the new fence.

5

[Turn over]



\* X 7 4 4 7 5 0 1 0 7 \*

7. The table below shows the vehicle tax to be paid on different vehicles.

The amount of vehicle tax paid depends on the CO<sub>2</sub> emissions of the vehicle and the fuel type.

**Standard Rates** – The following table contains the rates of vehicle tax for already registered cars, based on CO<sub>2</sub> emissions and fuel type.

|        |                                              | Petrol Car (Tax Class 48) and<br>Diesel Car (Tax Class 49) |               |                               |                                                     |                                |
|--------|----------------------------------------------|------------------------------------------------------------|---------------|-------------------------------|-----------------------------------------------------|--------------------------------|
|        |                                              | Non Direct Debit                                           |               | Direct Debit                  |                                                     |                                |
| Bands  | CO <sub>2</sub><br>emission figure<br>(g/km) | 12<br>months                                               | Six<br>months | Single<br>12 month<br>payment | Total<br>payable<br>by 12<br>monthly<br>instalments | Single<br>six month<br>payment |
| Band A | Up to 100                                    | £0                                                         | –             | –                             | –                                                   | –                              |
| Band B | 101 to 110                                   | £20                                                        | –             | £20                           | £21                                                 | –                              |
| Band C | 111 to 120                                   | £30                                                        | –             | £30                           | £31·50                                              | –                              |
| Band D | 121 to 130                                   | £110                                                       | £60·50        | £110                          | £115·50                                             | £57·75                         |
| Band E | 131 to 140                                   | £130                                                       | £71·50        | £130                          | £136·50                                             | £68·25                         |
| Band F | 141 to 150                                   | £145                                                       | £79·75        | £145                          | £152·25                                             | £76·13                         |
| Band G | 151 to 165                                   | £180                                                       | £99           | £180                          | £189                                                | £94·50                         |
| Band H | 166 to 175                                   | £205                                                       | £112·75       | £205                          | £215·25                                             | £107·63                        |
| Band I | 176 to 185                                   | £225                                                       | £123·75       | £225                          | £236·25                                             | £118·13                        |
| Band J | 186 to 200                                   | £265                                                       | £145·75       | £265                          | £278·25                                             | £139·13                        |
| Band K | 201 to 225                                   | £290                                                       | £159·50       | £290                          | £304·50                                             | £152·25                        |
| Band L | 226 to 255                                   | £490                                                       | £269·50       | £490                          | £514·50                                             | £257·25                        |
| Band M | Over 255                                     | £505                                                       | £277·75       | £505                          | £530·25                                             | £265·13                        |

MARKS

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Tom buys a **petrol** car which has a CO<sub>2</sub> emission figure of 142 g/km.

Tom decides to pay his vehicle tax by direct debit in two single six month payments.

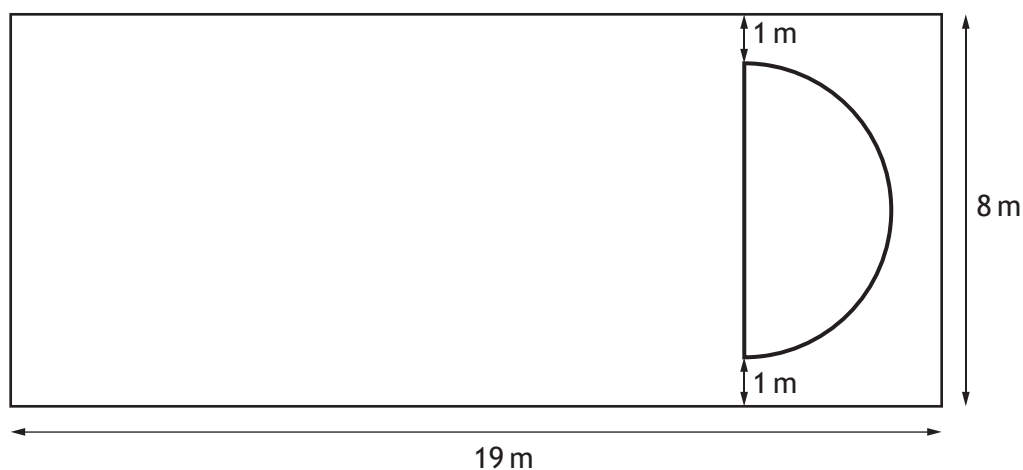
How much more expensive is this than a single 12 month payment by direct debit?

3



\* X 7 4 4 7 5 0 1 0 8 \*

8. A new playground is planned for Aberbeath Primary School.  
It will be a rectangle measuring 19 metres by 8 metres.  
A semi-circular sandpit will be built within the playground as shown



The playground, excluding the sandpit, is to be covered in rubber tiles.  
Calculate the area to be covered by the rubber tiles.

Take  $\pi = 3.14$ .

Give your answer to **3 significant figures**.

4

[Turn over



9. A picture is glued onto a piece of card as shown.



- The picture is a rectangle with dimensions 4 cm by 5 cm.
- The rectangular card has an **area** 2.8 times greater than the **area** of the picture.
- One of the dimensions of the piece of card is 7 cm.

Calculate the other dimension of the piece of card.

3



10. Bradley decides to cycle from Kilsyth to the highest point of Tak-Ma-Doon Road.

- The horizontal distance between these two places is 4.5 kilometres.
- Kilsyth is 70 metres above sea level.
- The highest point of Tak-Ma-Doon Road is 320 metres above sea level.

(a) Calculate the average gradient between Kilsyth and the highest point of Tak-Ma-Doon Road.

Give your answer as a fraction in its simplest form.

3

(b) One part of the road has gradient  $\frac{2}{25}$ .  
Is this steeper than the average gradient?

You must justify your answer.

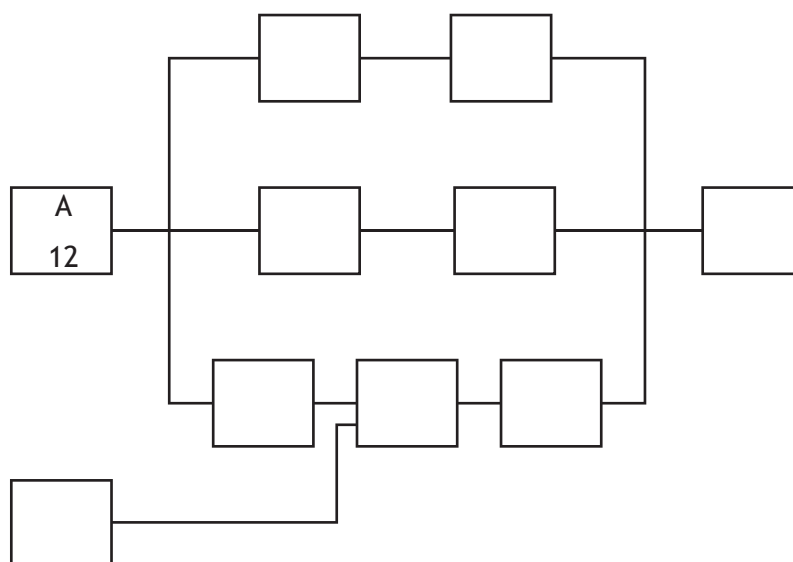
2

[END OF QUESTION PAPER]



## ADDITIONAL SPACE FOR ANSWERS

Additional diagram for Question 5 (a)



\* X 7 4 4 7 5 0 1 1 2 \*

MARKS

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ADDITIONAL SPACE FOR ANSWERS



\* X 7 4 4 7 5 0 1 1 3 \*

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ADDITIONAL SPACE FOR ANSWERS



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\* X 7 4 4 7 5 0 1 1 5 \*

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\* X 7 4 4 7 5 0 1 1 6 \*

## ACKNOWLEDGEMENTS

Question 3 – Bus timetable is adapted from ‘Stagecoach Dumfries-Edinburgh Bus Timetable.’  
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Question 9 – a40757/shutterstock.com



\* X 7 4 4 7 5 0 1 1 7 \*